

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using `wait()` and `waitpid()`. Compare the behavior of both functions.

Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

```
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#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t child[i];
    int i;

    // Create 3 child processes
    for (i = 0; i < 3; i++)
    {
        child[i] = fork();
        if (child[i] < 0)
        {
            printf("fork failed\n");
            return 1;
        }

        if (child[i] == 0)
        {
            printf("child %d started, PID = %d\n", i + 1, getpid());

            sleep(i + 1);

            printf("child %d completed, PID = %d\n", i + 1, getpid());

            return 0;
        }
    }

    // Parent process
    printf("\nParent PID = %d\n", getpid());

    // Using wait() to wait for any child
    printf("Parent using wait()...\n");
    wait(NULL);
    printf("One child completed using wait()\n");

    // Using waitpid() for remaining specific children
    for (i = 0; i < 3; i++)
    {
        if (child[i] > 0)
        {
            waitpid(child[i], NULL, 0);
            printf("child PID %d completed using waitpid()\n", child[i]);
        }
    }

    printf("\nAll child processes completed.\n");
    printf("No zombie processes remain.\n");

    return 0;
}
```

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```
nishanth@LAPTOP-CU1RFQML:~$ vi program4.c
nishanth@LAPTOP-CU1RFQML:~$ gcc program4.c -o program4
\nishanth@LAPTOP-CU1RFQML:~$ ./program4
Child 1 started, PID = 1330
Child 2 started, PID = 1331

Parent PID = 1329
Parent using wait()...
Child 3 started, PID = 1332
Child 1 completed, PID = 1330
One child completed using wait()
Child PID 1330 completed using waitpid()
Child 2 completed, PID = 1331
Child PID 1331 completed using waitpid()
Child 3 completed, PID = 1332
Child PID 1332 completed using waitpid()

All child processes completed.
No zombie processes remain.
```